

WEEK 1: INTEGRATIVE PHYSIOLOGY

GENERAL PRINCIPLES & ENERGY PRODUCTION IN MEDICAL PHYSIOLOGY

GENERAL PRINCIPLE

The Body as Organized "Solutions"

Average young adult male weight:

- 7% mineral
- 15% fat
- 18% protein
- 60% water

EXTRACELLULAR FLUID - *outside of the cell (internal sea)*

- 1/3 of total body water - 20% of body weight

- plasma = 5%
- interstitial = 15%

- Sodium chloride

- Where waste is discarded

- In animals with closed vascular system

- **Interstitial fluid** - part of the ECF that bathes the cell, takes up nutrients and oxygen, and releases waste products
 - Edema - abnormal volume interstitial fluid
- Blood plasma
- Lymph fluid - bridges the first two

- *Total blood volume = 8%*

INTRACELLULAR FLUID - *inside of the cell*

- 2/3 of total body water - 40% of body weight

- Potassium, phosphate, bicarbonate ions, protein

UNITS FOR MEASURING CONCENTRATION OF SOLUTES

1.) Moles (Standard Unit)

- 1 mol = 6×10^{23} molecules - Avogadro's number

- 1 mmol = 1/1000
- 1 qmol = 1/1,000,000

- Gram-molecular weight of a substance

- Molecular weight - weight of a single molecule

- **Da (Dalton)** - unit used to measure mass of protein

2.) Equivalents

- 1 eq = 1 mole of ionized substance / its valence

- Electrical equivalence is not = chemical equivalence

3.) Normality

- N = number of gram equivalent per liter of solution

- *Example:* 1 N HCl → contains H (1.0g)⁺ + Cl (35.5) = 36.5 g/L

Water, Electrolytes, & Acids/Bases

1.) H₂O

- Ideal solvent

- Has a dipole moment - oxygen pulls electrons from hydrogen to create charge separation to make it **polar** → **can easily pass through lipids**.
 - Allows substances (solute) to dissolve
 - Allows to interact to other H₂O molecule for hydrogen bonding
 - Water has a high surface tension
 - Water has a high heat vaporization and heat capacity
 - Water has a high dielectric constant

2.) ELECTROLYTES

- Ideal solute

- Dissociates in water and divided into charged particles called *ions*

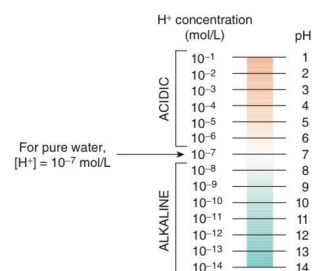
- **E.g. NaCl** - it breaks into Na and Cl when in water because of the water's net charge or polarity. It doesn't reassociate because the water molecules hugs each ion

- Important electrolytes: Na, K, Cl, Ca, Mg, HCO

- Uneven distribution of electrolytes are important for nerve signaling, and other physiological functions across the body.

- Necessary for resting membrane potential and action potential
- **E.g. K and Na** - K is more prominent inside of the cell, and Na outside of the cell. This separation calls for electrical gradient, important for membrane and action potentials to work

pH and Buffering



- pH = -log₁₀ [H]

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- **ACIDIC** = pH less than 7.0 = H is increased 10-fold
- **ALKALINE** = pH greater than 7.0 = H is decreased 10-fold

- Plasma of a person - slightly alkaline (7.35-7.45)

- Gastric fluid - acidic (3.0)
- Pancreatic secretion - alkaline (8.0)
- Pure water at 25°C - neutral (7.0)

- enzyme and protein are sensitive in pH

- **ACID** - H donors / **BASES** - H receivers

- Strong acids and bases dissociate in water
- Most physiological acids/bases are weak (partially dissociate)

- **Buffer** stabilizes or neutralizes pH by giving or removing H depending on the level of acid or base

- **Isohydric principle** - All buffers are in equilibrium (weak acid = conjugate base)
 - All buffers in the same solution share the same [H]

- **Equations:**

- Common dissociation equation:
 - $HA = H + A$
- Dissociation constant:
 - $K = [H][A] / [HA]$
- Henderson-Hasselbalch equation:
 - $pH = pK + \log[A] / [HA]$

Carbonic acid-bicarbonate buffer system

- $H_2CO_3 = H + HCO_3$
- Resists pH changes by shifting acids/bases

Diffusion

- Process by which a gas or substance expands in a solution, or moves from a region to another

- A particle is likely to move away from a high concentration area

- **Net flux** - overall movement of particles

- The time required for equilibrium depends on the distance

- Longer travel = slower diffusion

- Factors affecting diffusion:

- **Concentration gradient** - the higher the concentration, the more it will likely to travel
- **Cross-sectional area** - the larger the area, the more substance will be able to go through at once

- **Thickness of the boundary** - the thicker of the boundary, the slower the diffusion

Osmosis

- The movement of water molecule from an area of higher water concentration, going to higher solute concentration to balance the water molecule concentration

- **E.g. You have two cups:** One with pure water, and one with sugary water. The pure water would want to move to the sugary water since the water molecule concentration there is already too small because of the sugar concentration in it. Water would always want to be equal in terms of water molecule concentration.

- **Osmotic pressure** stops the water molecule to move to the higher solute concentration

- $P = nRT / V$
 - n - number of particles
 - R - gas constant
 - T - absolute temperature
 - V - volume
 - Unit: osmoles
- Colligative property - It depends on the number of solutes, not the type
- Membrane is important because if it is permeable to solute solutions, there would already be a balance, causing for osmotic pressure to not happen

Osmole (Osm)

- 1 Osm = gram molecular weight / number of freely moving particles per molecule in solution

- Milliosmoles (mOsm) - often used by biological solutions

- Only depends on molecule count

- NaCl -> $Na + Cl = 2$ Osm per mole
- 2NaSO₄ -> $2 Na + SO_4 = 3$ Osm per mole

- **Real body fluids vs ideal solutions**

- Ion interactions reduce the number of free particles
 - In body fluids: NaCl do not dissociate
 - In ideal solutions: NaCl breaks down into Na + Cl
- 1 mmol of NaCl in body fluids contributes less than 2mOsm

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- NaCl in body fluids do not break down. Thus, less than 2 mOsm
- Greater concentration = larger deviation
 - The more NaCl you put in water, the more likely they will interact
- Osmolarity (Solute) = number of osmoles per liter of solution
 - Affected by solute volume and temperature
- Osmolality (Solvent) = number of osmoles of solute kilogram of solvent (usually water)
 - Temperature dependent

Osmolal: Concentration of Plasma: Tonicity

- **Tonicity** is the osmolality relative to plasma
 - **Isotonic** - same osmolality as plasma
 - **Hypotonic** - lower osmolality (cell swell)
 - **Hypertonic** - higher osmolality (cell shrink)
 - Solutions initially isosmotic as plasma will remain isotonic, unless metabolizing and diffusion happens
 - **E.g. 0.9% saline solution** - it will remain isotonic since it will not be metabolized and absorbed by the cells because of its salt
 - **E.g. 5% glucose solution** - it is isotonic at first, but then the body will metabolize the glucose, absorbing it, causing the overall concentration to be reduced. This will cause hypotonicity, and the remaining concentration in the plasma will go inside the cell to balance things out
- Osmolality in plasma is made up mostly by bicarbonate, sodium, and chloride - 270 mOsm/L
 - Even though plasma protein is large, they are so big that they can't be broken down
- Major nonelectrolyte - urea and glucose - 5 mOsm/L
 - **Disease:** hyperglycemia, uremia
- Total plasma osmolality is important in assessing overhydration, dehydration, and other electrolyte abnormalities

Nonionic Diffusion

- Moving of undissociated (neutral) acids and bases inside of cell, and then dissociate inside

Donnan Effect

- Balances the concentration inside and outside of the cell
- **Charged particles (e.g. protein)** cannot pass through the membrane, causing ions and water to move across the membrane to balance things out. This causes the imbalance of ions and water concentration in each side, leading to a difference in osmotic pressure
 - Cell would swell
 - Prevented by Na/K ATPase pump, which maintains normal cell volume and pressure
- Gibbs-Donnan equation:
 - $[K][Cl] = [K][Cl]$

Forces Acting on Ions

1.) Concentration gradient

- Difference in concentration between the inside and outside of cell
- Naturally, ions move from *higher concentration areas* to *lower concentration areas*
- **E.g.** Chloride ions are more concentrated outside the cell, which will eventually move its way inside the cell (higher to lower)

2.) Electrical gradient

- Inside of the cell is negative and outside is positive
- Negativity of the inside attracts positive ions, making the negative pushed outside (e.g. chloride)

Equilibrium potential (Nernst potential)

- Concentration gradient = electrical gradient
- *No net movement of chloride ions* in the membrane
- **-70 mV** - equilibrium potential of chloride (same as RMP)

SUMMARY:

Chloride ions are more concentrated outside the cell, making it's way inside by concentration gradient (higher con to lower con). However, due to electrical gradient, the chloride ions' negativity and the inside of the cell's negativity, it is pushed back. The concentration and

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electrical gradient is happening back and forth, until such time when equilibrium potential is established (no net movement)

Establishment of Membrane Potential

Na/K ATPase Pump

- Site in the membrane, wherein lets 3 Na out and 2 K in the cell, powered by ATP

- Potassium ions:
 - EP = -90 mV
- Sodium ions:
 - EP = 60 mV

- *Stabilizes the equilibrium potential*

- Electrogenic - contributes to membrane potential

Membrane potential

- Electrical charge across the cell membrane
- Reason why there's electrical gradient
- Semipermeability of cell membrane causing it to have membrane potential

Energy Production

Energy Transfer

- Energy mostly came from phosphate groups
- When this bond is broken down, it releases a lot of energy (around 10-12 kcal/mol)

- **Hydrolysis** - breaking down of bonds
- **High-energy phosphate compounds** - compounds that stores lots of energy

- **Adenosine triphosphate (ATP)** - important and well-known energy source

- Energy storehouse of the body
- If it undergoes hydrolysis (loses 1 phosphate group), it turns into *ADP*
 - muscle contraction, active transport, synthesis of chemical compounds
- If it undergoes hydrolysis once again, turns into *AMP*
 - Releases more energy

Biological Oxidations

Oxidation

- Combination of a substance with O₂, or loss of hydrogen

- **Reduction** - reverse process of oxidation

- **Cofactors (ions) or coenzyme (organic molecules)** - carrier for product of oxidation, catalyze a variety of reaction

- Catalysis - acceleration of a chemical reaction by catalyst
- Coenzymes can catalyze multiple reactions

Oxidative phosphorylation

- Process by which ATP is formed in the body, specifically in mitochondria, using proton gradient

- 90% of basal O₂ consumption occurs in mitochondria; 80% of that is coupled to ATP synthesis

- Aerobic production (80%) - uses oxygen

- Anaerobic production (20%) - doesn't use oxygen

ATP utilization in the cell

- 27% - protein synthesis
- 24% - Na/K ATPase
- 9% - gluconeogenesis
- 6% - Ca ATPase
- 5% - myosin ATPase
- 3% - ureagenesis

Molecular Building Blocks

Nucleosides, Nucleotides, & Nucleic Acids

Nucleosides

- Has two parts: *sugar* (either ribose or 2-deoxyribose) and *nitrogen-containing base* (e.g. purines and pyrimidines)

- Sugar has ring structure

Nucleotide

- When inorganic phosphate is added to nucleoside
- Crucial for ATP production, RNA and DNA production
- Can be found in dietary nucleic acids (digested) or synthesis from amino acids (mainly in liver)

Nucleic Acid

- Can be consumed by the food we eat, which then breaks down into purine and pyrimidine - can be reused in building RNA and DNA

- DNA cannot change through lifetime
- RNA can be made or broken down

- If purine and pyrimidine is not used, it will be further broken down or reused, and might be excreted in urine

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Uric Acid

- Purines breaks down and creates this
- Healthy human: 4 mg/dL
- **Gout** - too much uric acid in blood and in urine

DNA

- Found in bacteria, nuclei of eukaryotic cells, and in mitochondria
- Contains long nucleotide chain - **Adenine, Thymine, Cytosine, Guanine (AT-CG)**
 - Bound together by hydrogen bonding
- Forms a stable double helix
- Compacted by histones -> chromosomes
 - 46 chromosomes (23 pairs) - diploid human cell
 - Have *two alleles*, one from each parent

Gene

- Fundamental unit of DNA
- Contains information for making ordered amino acid sequence
- Difference in alleles can alter gene properties
 - **Single nucleotide polymorphisms (SNPs)**
 - Single base changes with potential major functional effects
 - Important in human disease research

Gene mutation

- Gene is altered from its original DNA sequence
- Can be *through insertions, deletions, or duplications*
- **Point mutation** - only one pair (single base) is substituted
- Caused by *chemical (alkylating/intercalating agents) or ionizing radiation*

Genome

- Collection of gene in organism's DNA - contains blueprint for all heritable characteristics and proteins
- *One human haploid genome (1 chromosome) = 3 billion base pairs = 30,000 genes*

Gene expression

- All nucleated somatic cells have the full genome
- Only a small portion is transcribed in each cell type
- Genes regulated spatially (by location) and temporally (by time)
- Double helix must be precisely unraveled for replication or transcription

Replication: Mitosis & Meiosis

- *During mitosis*, the DNA double helix strand is separated into two strands. This is necessary to create a new copy of DNA

- The enzyme **DNA polymerase**, helps with the synthesis of the new strands
- New cell (daughter cell) gets a one full set of copy of DNA
- **Cell cycle** - cells' life
 - **G1** - cells growth and small portion of DNA synthesis
 - **S** - DNA synthesis (cell has two full set of DNA)
 - **G2** - further cell growth before mitosis
 - **M** - chromosome condensation and mitosis

- **Meiosis** - germ cell division

- Reductive division - reduces the chromosome number by half (62 -> 31)
- Each mature germ cell (egg and sperm) has one chromosome from each pair
 - 23 chromosome for sperm; 23 chromosome for egg
- Fertilization restores full chromosome complement in the zygote (half from each parent)
 - Two mature germ cell combine to summate back to 46 chromosome

- **Ploidy** - term used to determine the number of chromosomes in cell

- Diploid - normal resting diploid cells, turns into tetraploid
- Tetraploid - just before cell division
- Aneuploidy - abnormal number of chromosomes

RNA

- Single-stranded

- *Adenine, Uracil, Cytosine, Guanine*

- Sugar is ribose

- **Transcription** - process of *making RNA from DNA* -> done by enzyme RNA polymerase

- Messenger RNA (mRNA)
- Transfer RNA (tRNA)
- Ribosomal RNA (rRNA)

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TRANSCRIPTION AND TRANSLATION

RNA makes a copy through DNA via RNA polymerase -> the RNA copy is called mRNA -> mRNA carries message to make protein -> goes to cytoplasm and ribosome reads the message and dictates the formation of polypeptide chain -> tRNA brings amino acid necessary for making protein -> becomes rRNA

AMINO ACIDS & PROTEINS

Amino Acids

- Forms protein
- Can be *acidic, neutral, or basic*, depending on the relative proportions of basic or acidic group in molecule
- **Essential amino acids**: must be obtained from diet
- **Conditionally essential**: arginine and histidine - amino acid that should be consumed, doesn't grow within
- **Nonessential**: can be synthesized in the body

Metabolic Classifications:

- **Ketogenic amino acids** - transformed into acetoacetate (ketone body)
- **Glucogenic amino acids** - transformed into glucose

The Amino Acid Pool

- **Peptide** - 2-100 amino acids in molecule
 - *Polypeptide* - more than 10 but less than 100
- **Protein** - over 100

Protein

- building block up of amino acids, linked by peptide bonds
- Contains *carbohydrate* (glycoprotein) and *lipids* (lipoprotein)

Structure:

- Primary
- Secondary
- Tertiary
- Quaternary

Protein Synthesis

- RNA makes a copy through DNA via RNA polymerase -> the RNA copy is called mRNA -> mRNA carries message to make protein -> goes to cytoplasm and ribosome reads the message and dictates the formation of polypeptide chain -> amino acid is combined by the enzyme called AMP (adenylate) -> makes amino acid

ready -> tRNA brings amino acid necessary for making protein -> tRNA's anticodon attaches to mRNA's codon -> becomes rRNA

Carbohydrates

- Made up of carbon and H₂O
- **Sugar (glucose)** - building block of carbohydrate
 - Monosaccharides
 - Pentoses (ribose) - part of nucleotide
 - Hexoses (glucose) - help stores energy, main digestion product, and main circulating sugar
 - Disaccharides
 - Sucrose
 - Polysaccharides
 - Glycogen
- **Glucose, galactose, fructose (hexose polymers)** - sugars we eat
- **D isomers** - most body monosaccharides
- **Glycoprotein** - sugars on protein for cell targeting and signal recognition

Glucose enters cell -> converted to glucose-6-phosphate by the help of hexokinase (or glucokinase in liver) -> can be turned to glycogen or be broken down

- **Glycogenesis** - process forming glycogen from long chain of glucose (storage as glycogen)
- **Glycogenolysis** - process of breaking down glycogen into glucose
- **Glycolysis** - process of breaking down pyruvate or lactate -> turning into glucose -> turned into acetyl-CoA, used for citric acid cycle
- **Gluconeogenesis** - lactate or nonglucose molecules to glucose
 - Glucose to fats (via acetyl-CoA)
 - Fats to glucose - only small contribution from glycerol because acetyl-CoA conversion to pyruvate is irreversible

FATTY ACIDS AND LIPIDS

- **Main types**: fatty acids & derivative, neutral fats (triglycerides), phospholipid, sterols
- **Triglycerides** - 3 fatty acids + glycerol
- Fatty acids are broken down to acetyl-CoA to enter citric acid cycle

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- Break down in mitochondria through a process called **β -oxidation**
 - **Saturated** - no double bonds
 - **Unsaturated** - 1+ double bonds
- **Phospholipids** - main component of cell membrane
- Source of intra- and intercellular signaling molecules
 - Fatty acids = major energy source in the body

- **Statin** - drugs that lowers cholesterol by blocking HMG-CoA reductase

Cellular Lipids

TWO TYPES OF LIPIDS IN CELLS:

1.) Structural lipids - part of cell membrane, helps cell to communicate, preserved during starvation

2.) Neutral fat - fat stored in the adipose tissue, then mobilized during starvation, broken down for energy consumption -> 15% men; 21% in women

- Glucose -> fatty acid -> neutral fat

3.) Brown fat - more abundant in infants or newborn, burns calories to produce heat

- Only small % of total body fat
- Found between scapula, nape of neck, around major abdominal/thoracic vessels, other scattered sites
- Extensive sympathetic innervation of fat cells and blood vessels

Plasma Lipids and Lipid Transport

- Major lipids are water-insoluble; do not circulate freely

- **FFAs** - bound to albumin

- **Cholesterol, triglycerides, phospholipids** - transported as lipoprotein complexes

- Lipoproteins: hydrophobic core (triglycerides, cholesteryl esters) + outer layer (phospholipid, proteins)

- Six lipoprotein families

- HDL, LDL, VLDL, IDL, chylomicron, chylomicron remnants
- Size and lipid content varies
- Density is inversely proportional to lipid content
-> more lipid, less dense (vice versa)

- **HDL** - more protein

- **Chylomicron, LDL, VLDL, & IDL** - cholesterol

Cholesterol Metabolism

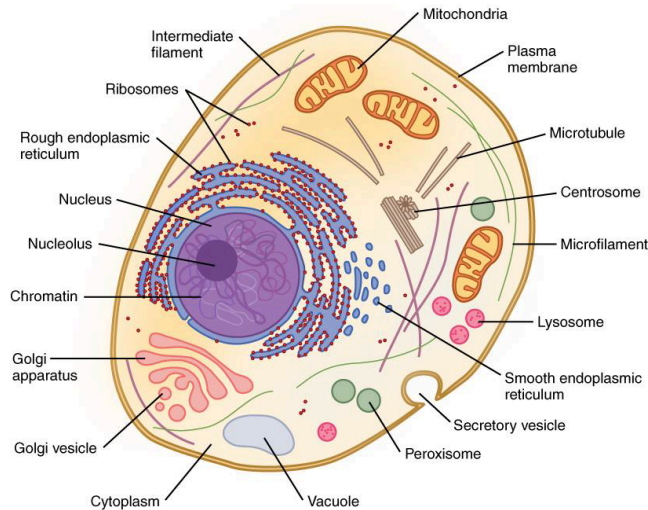
- **Cholesterol** - *building block of steroid hormone*

- Makes bile acids
- Made from acetate (basic building block)

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OVERVIEW OF CELLULAR PHYSIOLOGY

Functional Morphology of the Cell and Homeostasis



- Extracellular fluid dictates a normal cell to function
- **Homeostasis** - maintain the internal balance in our body
- Functions:
 - Smallest unit of life
 - Cell metabolism and energy use
 - Synthesis of molecules
 - Communication
 - Reproduction and inheritance

Cytoplasm

- Jelly-like substance that holds organelles

Cell Membrane

- **Acts as a barrier**, controls what enters or exits the cell (semipermeable)
- Also referred to as *plasma membrane*
- *Made up of lipids and proteins*
- Main component is phospholipid
 - **Amphipathic** - what makes it semipermeable
 - Head is polar and hydrophilic - structure is exposed outside where there is water
 - Tail is non polar and hydrophobic - structure is exposed inside where there is no water
- **Proteins embedded:**
 - Integral proteins (intrinsic) - passes through the cell membrane
 - Peripheral proteins (extrinsic) - only surface
- Inside of cell - K, glycogen, enzymes, proteins

- Outside of cell - Na, Cl, Ca

Mitochondria

- Contains outer and inner membrane, intermembrane space, and matrix
- **Regulates apoptosis** (programmed cell death) and **oxidative phosphorylation** (located inside cristae), an energy-rich compound *ATP*
- Have own genome (less than nucleus) - 16,500 base pairs
- 99% of mitochondrial proteins are encoded by nuclear DNA
- **Inherited maternally from the ovum**

Lysosomes

- Large, irregular, membrane-bound structures in the cytoplasm
- Hydrolytic enzymes located inside **digest endocytosed bacteria** (bacteria that is brought into cell) and **worn-out cell components**
- The interior is more acidic than cytoplasm (pH 5.0), caused by the proton pump or H ATPase. Thus, function optimally at acidic pH
 - If it ruptures, enzymes are largely inactive in the neutral cytosolic pH (7.2)
- Vesicles formed by endocytosis maybe fused with lysosomes in order to breakdown materials
 - WBC phagocytizing bacteria

Peroxisomes

- Forms by budding from endoplasmic reticulum
- **Metabolize lipid and detoxify harmful substances**
- Contains enzyme that could either oxidases or catalases
- Membrane has peroxisome-specific transport proteins
- **Peroxisins** - chaperones that facilitates proteins' proper folding and transport to peroxisome

Cytoskeleton

- System of fibers that **maintains cell structure**, changes its shape, making it move, and supports intracellular transport
- **Parts:**
 - *Microtubules*

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- Tracks for organelle/vesicle transport
- **Forms flagella, cilia, and centrioles**
- Forms mitotic spindle for chromosome movement in mitosis
- Bidirectional cargo transport possible
- Temperature sensitive when growing
- **Intermediate filament**
 - 8-14 nm
 - Bigger than microfilament
 - **Connects nuclear membrane to cell membrane**; provide flexible scaffolding & resist mechanical stress
 - Form external scaffolding for the cell and help it resist external pressure
 - Cell markers:
 - **Vimentin** - fibroblast
 - **Cytokeratin** - epithelial cells
- **Microfilaments (actin filament)**
 - Attach to cytoskeleton parts and membrane proteins
 - Form lamellipodia for cell crawling
 - Serve as tracks for some molecular motors
 - Made up of actin monomers
 - **Treadmilling** - can polymerize (attach) or depolymerize (detach) - occurs at opposite ends of filament
 - This allows movement, cell shape maintenance, endocytosis (engulfing), cytokinesis (cell division)

Centrosomes

- Found near the nucleus in cytoplasm of animal cells
- **Composed of two centrioles** + surrounding pericentriolar material
 - Centrioles - short cylinders, arranged at right angles
 - **Has 9 triplets** of microtubules spaced around circumference
- Also known as microtubule-organizing center (MTOC) - *help the formation of microtubules -> mitotic spindle for cell division*

Cilia

- Used by unicellular organisms to **propel themselves through liquid by mucus**
 - It beats to produce locomotion
- *All cells have primary cilium* - sensory organelle that receives mechanical and chemical signal
- Increase the surface area of cell
- Functionally similar to eukaryotic flagella (e.g., sperm tails)

Cell Adhesion Molecule

- Anchor cells to the basal lamina and to each other
- Crucial in embryonic development, nervous system formation, tissue cohesion, inflammation, wound healing, and tumor metastasis
- Many pass through cell membrane and anchor to the cytoskeleton inside the cell
- Binding types:
 - **Homophilic** - identical molecules to other cells
 - **Heterophilic** - to different molecules
- Many bind to laminins in the extracellular matrix
- Transmit signals into and out of the cell
- Loss of integrin contact with ECM increases apoptosis
- Integrin-cytoskeleton interactions aid cell movement
- **Major families of CAMs:**
 - Integrins - heterodimers binding to various receptors
 - IgG superfamily adhesion molecules - related to immunoglobulins
 - Cadherins - Ca-dependent, mediate cell-to-cell adhesion via homophilic binding
 - Selectins - have lectin-like domains binding to carbohydrates

Intracellular Connection

- Fasten cells to each other and surrounding tissue - provide strength and stability
- Permit transfer of ions/molecules - allow intercellular communication

Tight Junctions (Zonula Occludens)

- Found at apical margins of epithelial cells (e.g., intestinal mucosa, renal tubules, choroid plexus)
- Important for endothelial barrier function and vasodilation
- Made of occludin and claudins

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- Seal space between cells, controlling paracellular transport (leakiness varies by protein makeup)
- Prevent lateral movement of membrane proteins, maintaining polarized distribution of transporters/channels

Zonula adherens

- Located just basal to tight junctions
- Major site for microfilament attachment
- Contain cadherins

Focal adhesions

- Attach cells to basal lamina
- Associated with actin filaments
- Labile, dynamic structures important in cell movement

Desmosomes and Hemidesmosomes

Desmosomes

- Patch-like junctions with membrane thickening in adjacent cells
- Intermediate filaments attach internally; **cadherins** present in intercellular space

Hemidesmosomes

- Half-desmosomes that attach to the basal lamina
- Contain **integrins** instead of cadherins
- Connected to intermediate filaments

Gap Junctions

- Cytoplasmic "tunnels" connecting neighboring cells
- Permits passage of ions, sugars, amino acids, and other small solutes (< 1000 Da)
- Enables rapid electrical activity propagation and exchange of chemical messengers/signaling molecules
- Allow regulated small molecule passage, enabling extra signaling pathways

Nucleus and Relation Structures

- Present in all eukaryotic cell
- Contains chromosomes - carries the blueprint of a specie or individual
 - Occurs in pairs (except in germ cells)
 - Each is made up of one *giant molecule of DNA*
 - **Genes is located** - ultimate unit of heredity
- **Parts:**

- **Nuclear membrane or envelope**

- Double membrane enclosing nucleus; space between layers (inside) = perinuclear cisterns
- Permeable to small molecules only

- **Nucleosome** - histone proteins (beads) wrapped with DNA strand (string) with linker DNA (interval in string)

- *Altogether*, called **chromatin**

- **Nucleolus**

- RNA-rich granular structure; sometimes multiple per nucleus
- Site of ribosome synthesis

DNA strand -> nucleosome is created via wrapping of histone protein (beads) -> linker DNA is the string of interval -> chromatin is the overall structure

Endoplasmic Reticulum

- Series of tubules in the cytoplasm of the cell

- **3 types:**

- **Rough endoplasmic reticulum (granular)** - ribosome is attached to the cytoplasmic side of the membrane.
 - Thus, protein synthesis
 - Initial folding of polypeptides, formation of disulfide bonds
- **Smooth endoplasmic reticulum (agranular)** - ribosomes are absent.
 - Thus, steroid synthesis in steroid-secreting cells and site of detoxification in other cells
- **Sarcoplasmic reticulum (modified ER in muscle cells)** - skeletal, cardiac, smooth muscle.
 - Contains intracellular Ca
 - Releases Ca for signaling - triggers contraction, proliferation, and migration

Ribosomes

- **Site of protein synthesis**

- Found in cytoplasm
- 3 RNA: mRNA, tRNA, rRNA
- Attached ribosomes (bound to RER):

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- Synthesize transmembrane proteins, most secreted proteins, and proteins stored in golgi, lysosomes, and endosomes
- Free ribosomes (in cytoplasm):
 - Synthesize cytoplasmic proteins (e.g., hemoglobin) and proteins for peroxisomes & mitochondria

Golgi Apparatus & Vesicular Traffic

- Present in the eukaryotic cell
- **Sorts, modifies, and packages molecules for delivery in or out of the cell**
- Polarized with cis face (near ER) and trans face (near membrane)
- **Glycosylation** - modifying proteins and lipids by adding sugar molecules to them, and it can go either out or in the cell
- How it works:
 - Proteins made from the RER is packaged into tiny bubbles (vesicle) and travel out the cis face of golgi (endoplasmic reticulum)
 - Inside the golgi, the proteins are modified and goes out the trans face of the golgi (plasma membrane)
 - From trans face, the vesicles will either travel to lysosomes or the cell exterior via constitutive or regulated exocytosis, or be recycled via endocytosis
- Vesicular traffic regulation
 - Small G-proteins (GTP/GDP binding): control vesicle assembly & delivery
 - SNARE proteins:
 - v-SNAREs (vesicle) bind specifically to t-SNAREs (target membrane) in lock-and-key manner
 - Membrane proteins and lipids on vesicles helps target them to corrects compartments

Apoptosis

- Also known as **programmed cell death (cell suicide)**
 - Cell dies and absorbed under genetic control
- Different from necrosis
 - “Cell murder” due to external damage like inflammation - it swells and ruptures

- Results to DNA fragmentation, cytoplasmic & chromatin condensation, and membrane blebbing (cell breakup causes debris removal by phagocytosis)

TRANSPORT ACROSS CELL MEMBRANE

Active and Passive Membrane Transport

- **Active transport** - requires ATP to move from low to high concentration and vice versa
- **Passive Transport** - through diffusion, osmosis, and facilitated diffusion

Exocytosis

- Vesicle is prepared for export to go out of the cell membrane
 - **v-SNARE**- protein within the vesicle
 - **t-SNARE** - protein on the cell membrane
 - Locks together in order for the vesicle to be ready for transport
- Membrane fusion releases vesicle contents outside the cell while keep the plasma membrane intact
- Calcium-dependent

Endocytosis

- **Brings material into itself** (as opposed to exocytosis where materials are transported outside)
- **Types:**
 - *Phagocytosis (cell eating)*
 - Eating or engulfing bacteria and dead cells by immune cells like polymorphonuclear leukocytes
 - Forms a large membrane-enclosed vacuole
 - *Pinocytosis (cell drinking)*
 - Uptake of fluid and dissolved substances in small vesicle
 - Can internalize the equivalent of the entire cell membrane within ~1 hour (e.g., macrophages)

In summary, exocytosis and endocytosis work together to add or remove surface area within the cell membrane

Membrane Permeability & Membrane Transport Proteins

WEEK 1: INTEGRATIVE PHYSIOLOGY

Passive membrane permeability

- Easily permeable: small non polar molecules (O₂, N₂) & small uncharged polar molecules (CO₂)
- Very limited permeability for most other substances -> require endocytosis, exocytosis, or transport proteins
- Water transport: partly by diffusion, mostly via aquaporins

Facilitated diffusion

- Carrier protein does not require energy to move to electrical or chemical gradient (e.g., glucose transport from ECF to cytoplasm)
- Move molecules against gradients -> active transport (requires energy from ATP hydrolysis)
- **Example of ATP-powered pump**
 - *Sodium-potassium pump (Na/K ATPase)* - Na out, K in
 - *Proton pump (H/K ATPase)* - H help in creating stomach acid, present in lysosome and golgi
 - *Calcium pump (Ca ATPase)* - Ca out
 - *Proton ATPase* - acidify Golgi & lysosomes

Mode of transport

- Uniport - transports one substance
- Symport - cotransports is greater than or equal to substances in the **same direction** (e.g., Na-glucose symport in intestinal mucosa)
- Antiport - exchanges one substance for another **opposite direction**

- What can pass through?

- Small, non-polar (O₂, N₂) - can diffuse easily
- Small, polar (CO₂) - can diffuse but not easily
- *Larger substances* can pass through by endo/exocytosis, transport proteins, ion channels, aquaporins (for water)
 - Ion channels lets specific ions only
- Carrier protein allows molecules to move from one side of the membrane to the other
 - Changes it shape and bind to the molecules
 - Can be either uniports (transport one substance), or symports (more than one)
- **Movement along gradients**
 - *Chemical or concentration gradient* - naturally moves from high to low concentration
 - *Electrical gradient* - movement of cations to negative charged areas, anion to positive charged area

Ion channels

- Includes K, Na, Ca, and Cl
- **Gating mechanisms:**
 - *Voltage-gated channels*
 - Uses membrane potential in order to open or close
 - *Ligand-voltage channels*
 - Uses ligand to open or close
 - Neurotransmitter, hormones
 - *Mechanosensitive channels*
 - Sensitive to physical stretching, important for cell movement and responsiveness

Active Transport

- Carrier-mediated process, requiring ATP, that moves substances from lower to higher concentration against concentration gradient

Na, K ATPase

- Major example of active transport
- Catalyzes hydrolysis in ATP, causing it to become ADP which uses this energy to transport 3 Na out, and 2 K in the cell (3:2)

Secondary active transport

- Doesn't require ATP
- Uses energy provided by a concentration gradient established by the active transport of one substance, such as Na to transport other substances
 - *In cotransport*, the diffusing substance moves in the same direction as the initial active transported substance
 - *In countertransport*, the diffusing substance moves in a direction opposite to that of the initial transported substance

Transcytosis

- Coated with caveolin - necessary for vesicle membrane formation
- Endothelial cell transports protein across capillary walls
- *Connected to exocytosis and endocytosis*

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- Basically, takes something from a structure and then sending it to a different structure (or opposite side)

Intercellular Communication

- Cells communicate using chemical messengers via various pathways, triggering intracellular changes through receptor binding (surface, cytoplasmic, or nuclear)

Kinds:

- Gap junctions - allow messengers to pass directly between cells without entering the ECF
- Juxtacrine communication - requires direct membrane contact, going directly to the cell
- Neural communication - neurotransmitters released at synaptic junctions, acting across narrow clefts on postsynaptic cells

Types:

- Endocrine communication - hormones/growth factors travel through blood or lymph to distant target cells
- Paracrine communication - cell products diffuse locally in ECF to affect nearby cells
- Autocrine communication - cells secrete messengers that act on their own receptors

Receptor:

- Down regulation - decrease in active receptors when there are lots of excess hormone/neurotransmitter
- Upregulation - increase active receptors (hormone deficient)
 - Exception - Angiotensin II increases the receptor numbers in the adrenal cortex

Mechanism by which Chemical Messengers Act

- Chemical messenger binds to its receptors to trigger intracellular signaling

- Four main categories of responses:

- Ion channel activation - e.g., acetylcholine directly changes membrane conductance
- G-protein activation - triggers downstream signaling molecules
- Enzyme activation - receptor activates intracellular enzymes
- Direct transcription activation - e.g., steroid, thyroid hormones bind to nuclear/cytoplasmic receptors to regulate gene expression

First vs. Second Messengers

- First messengers - extracellular ligands (e.g., hormones, neurotransmitters)
- Second messengers - intracellular mediators (e.g., cAMP, IP3, DAG) that amplify and distribute signal
 - Short-term effects: alter enzyme activity, trigger exocytosis
 - Long term effects: alter transcription of genes

Intercellular Ca as a Second Messenger

- It regulates diverse processes: proliferation, neural signaling, learning, contraction, secretion, fertilization
- Intracellular storage - stored in high amounts in endoplasmic reticulum (ER) and other organelles
- Released via ryanodine and IP3 receptors

G-Proteins

- Made up of alpha, beta, and gamma subunits: alpha and beta have lipid that anchors to plasma membrane
- **G-Protein-Coupled Receptors (GPCRs)**
 - Structure: 7 transmembrane helices (serpentine receptors)
 - Amplification: one receptor can activate multiple G-proteins
 - Importance: GPCRs are major drug targets due to diversity and central role in signaling

Inositol Triphosphate (IP3) and Diacylglycerol (DAG) as Second Messengers

- Link ligand, binding at membrane receptors to increase cytoplasmic Ca
- IP actions:
 - Diffuses to ER
 - Binds IP receptor (ligand-gated Ca channel)
 - Triggers release of stored Ca into cytoplasm
- DAG actions:
 - Remains in plasma membrane
 - Activates protein kinase C (PKC) isoform

Cyclic AMP (cAMP)

- Main actions of cAMP
 - Activates protein kinase A (PKA)

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- PKA phosphorylates proteins, which changes their conformation and activity

Cyclic GMP (eGMP) - Guanosine Monosphosphate

- Important in vision (rod and cone cells)
- Regulates eGMP-gated ions channels
- Activates eGMP-dependent kinase for multiple physiological effects